SCI-NOI v0.1

SCI-NOI v0.1 r0.2 Engineering Conformance Specification

Implementation-blind Stage B draft

SCI-NOI_ENGINEERING_CONFORMANCE v0.1/draft-r0.2

Not owner-accepted or frozen. No implementation conformity, validation, calibration, physical-noise, covariance-completeness, significance, performance, readiness, or production claim.

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Document identity: SCI-NOI_ENGINEERING_CONFORMANCE v0.1/draft-r0.2

Status: implementation-blind Stage B draft; not owner-accepted or frozen.

Normative authority: the six ordered modules in SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.2, binding-file SHA-256 5c6f954b457a546abcf74a1dc6dae190f2b22ea43c14edf34d2e4d2a8a704268. This ECS defines prospective evidence procedures only. It does not add or modify science and makes no present conformity claim.

1. Evidence model

Evidence for every requirement shall identify the exact product, method, parent, plan, generation, lifecycle state, and named use. Review shall compare exact serialized scientific identities and typed states, not filenames, shapes, generic success flags, approximate WCS matches, or undocumented defaults. Failure to establish a required fact yields unavailable or failed, as prescribed by the bound module.

2. GEN operator evidence

For each ordinary member, evidence shall expose the exact retained PTC occurrence identity, detector/channel identity, network membership, coefficient family/value/QC, projection contribution, unsigned denominator, coverage state, WCS/support, response state, assignment, and application count. A symbolic or fixture-level audit shall show that the modifier occurs exactly once and only on Z_i^PTC, with the denominator and every other MAP gate unchanged. The corresponding all-+1 result shall retain ordinary MAP identity. Missing required response shall be reported as unavailable without changing signal membership.

3. Population and finite-design evidence

Until the owner accepts the complete candidate, conformance for detailed finite-design mechanics and base product scope is not evaluable. Evidence may record candidate facts but shall label them proposed. After acceptance, evidence shall enumerate the complete coherence population; active positive- mass subset; zero-mass detectors; zero-total strata; singleton active strata; ambiguous memberships; canonical order; exact rational tolerance; admissible set; base candidate law; retry cap; accepted draw index; replacement and duplicate rules; complement-orbit treatment; weights; counts; ranks; generator/version; key bytes; namespace; and product scope.

No check shall evaluate an imbalance ratio on a zero denominator. If bounded first-accepted rejection is accepted, a finite-enumeration or exact argument shall demonstrate that, conditional on resolution, its output law is uniform over the admissible vectors under the stated symmetric base law. The review shall verify that complement-related assignments are distinct and that no complement pair is forced.

4. Completion and retry evidence

Evidence shall distinguish candidate rejection from admitted-member failure, and scientific replacement from an idempotent execution retry. The ensemble record shall reconcile N_requested, N_resolved, N_completed, N_admitted, N_unique_sign, and N_unique_orbit. Every admitted member shall have a successful terminal producer state before initial UNC admission. A retry retaining scientific identity shall show identical inputs and deterministic work plus a distinct I_attempt. Any changed assignment or scientific state shall have a new scientific identity.

5. UNC evidence

Evidence shall bind one exact all-member domain and reproduce the declared design-weighted zero-centered second moment without finite-sample mean subtraction or a B-1 divisor. Units shall be squared signal units. The record shall keep member counts, uniqueness counts, r_sign, r_map, effective Monte Carlo information, and numerical estimator uncertainty separate. Estimator uncertainty shall be explicit or typed unavailable.

The evidence shall label the product only as the conditional detector-sign- randomization marginal second moment/variance for the fixed declared target. It shall expose source/structured-residual imprint limitations and shall not claim repeated physical-noise variance, total MAP uncertainty, covariance completeness, precision, significance, or calibration.

6. Reciprocal and STD evidence

The proposed reciprocal successor is not evaluable before its owner decision and immutable profile/source rebinding. If accepted, evidence shall restrict it to the exact finite strictly positive parent domain and show that no consumer-weight, precision, exposure, support, validity, or coefficient role was attached.

STD evidence shall identify the exact immutable m_MAP numerator and compatible sigma_cond parent; reproduce zeta_cond; verify unit 1; and emit only the bounded standardized-MAP claim. It shall demonstrate that no interpolation, domain extension, implicit reciprocal, JINC substitution, or statistical-significance language was introduced.

7. Persistence, transform, and profile evidence

Persistence evidence shall identify the selected owner-approved mode and its requested/effective/resolved/applied/realized states, reconstructibility, numerical regeneration promise, retained sufficiency, unavailable later questions, completion, and failure behavior. A composite shall have its own plan; fallback shall not be inferred.

Transform, Wiener, and FRUIT evidence shall bind exact external owner identity and parity for every declared operator fact. Any learning, selection, continuation, update, or member-specific replay shall create required new generations and shall not enter a fixed ensemble.

Profile evidence shall preserve separately named request, applicability, eligibility, and realization facts and shall project exactly one action only from the exact positive conjunction. The r0.18 profile bytes remain immutable. Because r0.2 changes notation/action spelling, affected evaluation is unavailable until a new immutable profile/Registry binding exists.

8. Traceability and review result

REQUIREMENT_PREDICTION_TRACEABILITY.csv is the exact one-row-per-requirement crosswalk. Its authority codes resolve only to the 17 admitted scientific objects or the explicitly authorized process-only source/profile binding records. A conformance review shall report each row as pass, fail, unavailable, or not applicable with evidence identity. This draft supplies no such review result and establishes no conformity, validation, calibration, performance, readiness, freeze, or production status.

Appendix A. Exact Requirement/Prediction Traceability

SCI-NOI-REQ-001

Predictions: SCI-NOI-PRED-011 Authorities: SCOPE|TAX|OWNER ECS section: 1 Review: identity-and-role review

SCI-NOI-REQ-002

Predictions: SCI-NOI-PRED-004 Authorities: PTCB|MAPB|GRAPH|OWNER ECS section: 2 Review: operator identity review

SCI-NOI-REQ-003

Predictions: SCI-NOI-PRED-010 Authorities: GRAPH|FILTER|OWNER ECS section: 7 Review: fixed-relearned generation review

SCI-NOI-REQ-004

Predictions: SCI-NOI-PRED-002 Authorities: DESIGN|OWNER ECS section: 2 Review: coherence occurrence audit

SCI-NOI-REQ-005

Predictions: SCI-NOI-PRED-001 Authorities: DESIGN|OWNER ECS section: 3 Review: coefficient-mass stratum audit

SCI-NOI-REQ-006

Predictions: SCI-NOI-PRED-015 Authorities: SCOPE|OWNER ECS section: 3 Review: typed pending-scope review

SCI-NOI-REQ-007

Predictions: SCI-NOI-PRED-004 Authorities: MAPB|PTCB|OWNER ECS section: 2 Review: symbolic sign-placement audit

SCI-NOI-REQ-008

Predictions: SCI-NOI-PRED-004 Authorities: MAPB|PTCB|OWNER ECS section: 2 Review: inline-materialized parity review

SCI-NOI-REQ-009

Predictions: SCI-NOI-PRED-012 Authorities: MAPB|GRAPH|OWNER ECS section: 2 Review: response occurrence parity review

SCI-NOI-REQ-010

Predictions: SCI-NOI-PRED-015 Authorities: DESIGN|OWNER ECS section: 3 Review: population-state and denominator audit

SCI-NOI-REQ-011

Predictions: SCI-NOI-PRED-016 Authorities: DESIGN|OWNER ECS section: 3 Review: typed pending-disposition review

SCI-NOI-REQ-012

Predictions: SCI-NOI-PRED-014 Authorities: DESIGN|OWNER ECS section: 3 Review: owner-question gate review

SCI-NOI-REQ-013

Predictions: SCI-NOI-PRED-013 Authorities: DESIGN|LIFE|OWNER ECS section: 3 Review: plan preregistration identity review

SCI-NOI-REQ-014

Predictions: SCI-NOI-PRED-014 Authorities: DESIGN|OWNER ECS section: 3 Review: conditional-law proof after approval

SCI-NOI-REQ-015

Predictions: SCI-NOI-PRED-005 Authorities: DESIGN|OWNER ECS section: 3 Review: duplicate and complement audit

SCI-NOI-REQ-016

Predictions: SCI-NOI-PRED-005 Authorities: DESIGN|UNCT|OWNER ECS section: 4 Review: count rank and information reconciliation

SCI-NOI-REQ-017

Predictions: SCI-NOI-PRED-003|SCI-NOI-PRED-007 Authorities: DESIGN|OWNER ECS section: 3 Review: claim and source-imprint review

SCI-NOI-REQ-018

Predictions: SCI-NOI-PRED-008 Authorities: DESIGN|LIFE|OWNER ECS section: 4 Review: atomic completion audit

SCI-NOI-REQ-019

Predictions: SCI-NOI-PRED-013 Authorities: LIFE|OWNER ECS section: 4 Review: retry-versus-replacement identity audit

SCI-NOI-REQ-020

Predictions: SCI-NOI-PRED-009 Authorities: LIFE|OWNER ECS section: 7 Review: persistence plan review

SCI-NOI-REQ-021

Predictions: SCI-NOI-PRED-010 Authorities: FILTER|OWNER ECS section: 7 Review: external transform parity review

SCI-NOI-REQ-022

Predictions: SCI-NOI-PRED-010 Authorities: FILTER|LIFE|OWNER ECS section: 7 Review: generation-separation review

SCI-NOI-REQ-023

Predictions: SCI-NOI-PRED-006|SCI-NOI-PRED-007 Authorities: UNCT|OWNER ECS section: 5 Review: UNC contract completeness review

SCI-NOI-REQ-024

Predictions: SCI-NOI-PRED-008 Authorities: UNCT|VALP|OWNER ECS section: 5 Review: all-member admission audit

SCI-NOI-REQ-025

Predictions: SCI-NOI-PRED-002|SCI-NOI-PRED-006 Authorities: UNCT|OWNER ECS section: 5 Review: estimator reproduction

SCI-NOI-REQ-026

Predictions: SCI-NOI-PRED-007|SCI-NOI-PRED-009 Authorities: UNCT|OWNER ECS section: 5 Review: unit target and claim review

SCI-NOI-REQ-027

Predictions: SCI-NOI-PRED-009 Authorities: UNCT|OWNER ECS section: 5 Review: uncertainty-count-rank separation review

SCI-NOI-REQ-028

Predictions: SCI-NOI-PRED-005 Authorities: UNCT|TAX|OWNER ECS section: 5 Review: product-role separation review

SCI-NOI-REQ-029

Predictions: SCI-NOI-PRED-006 Authorities: UNCT|STDT|OWNER ECS section: 6 Review: typed reciprocal gate and domain review

SCI-NOI-REQ-030

Predictions: SCI-NOI-PRED-017 Authorities: STDT|JINCB|OWNER ECS section: 6 Review: numerator-scale-domain audit

SCI-NOI-REQ-031

Predictions: SCI-NOI-PRED-017 Authorities: STDT|OWNER ECS section: 6 Review: unit and claim-ceiling review

SCI-NOI-REQ-032

Predictions: SCI-NOI-PRED-011 Authorities: VALP|OWNER|REG_BIND ECS section: 7 Review: profile field/action projection review

SCI-NOI-REQ-033

Predictions: SCI-NOI-PRED-011 Authorities: VALP|OWNER|REG_BIND ECS section: 7 Review: GEN-input named-use audit

SCI-NOI-REQ-034

Predictions: SCI-NOI-PRED-008|SCI-NOI-PRED-011 Authorities: VALP|OWNER|REG_BIND ECS section: 7 Review: UNC-member named-use audit

SCI-NOI-REQ-035

Predictions: SCI-NOI-PRED-008|SCI-NOI-PRED-011 Authorities: VALP|OWNER|REG_BIND ECS section: 7 Review: UNC-ensemble named-use audit

SCI-NOI-REQ-036

Predictions: SCI-NOI-PRED-011|SCI-NOI-PRED-017 Authorities: VALP|STDT|OWNER|REG_BIND ECS section: 7 Review: STD successor-binding gate review

SCI-NOI-REQ-037

Predictions: SCI-NOI-PRED-012 Authorities: CONV|OWNER ECS section: 1 Review: typed-state and no-default review

SCI-NOI-REQ-038

Predictions: SCI-NOI-PRED-010 Authorities: LIFE|FILTER|OWNER ECS section: 7 Review: immutable-parent ancestry review

SCI-NOI-REQ-039

Predictions: SCI-NOI-PRED-009 Authorities: TAX|UNCT|STDT|OWNER ECS section: 6 Review: role-promotion prohibition review

SCI-NOI-REQ-040

Predictions: SCI-NOI-PRED-011 Authorities: VALP|OWNER|REG_BIND ECS section: 7 Review: profile-byte immutability review

SCI-NOI-REQ-041

Predictions: SCI-NOI-PRED-001 Authorities: COVER|OWNER ECS section: 8 Review: module hash and document binding review

SCI-NOI-REQ-042

Predictions: SCI-NOI-PRED-017 Authorities: SCOPE|COVER|OWNER ECS section: 8 Review: claim-ceiling language review